MIL-C-46221A(MU)
6 October 1966
SUPERSEDING
MIL-C-46221(ORD)
26 May 1960

MILITARY SPECIFICATION
CARTRIDGE, AIRCRAFT EJECTION SEAT, CATAPULT, M37 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Aircraft Ejection Seat, Catapult, M37 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS
Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Coatings, Anodic, Conventional, for Aluminum and Aluminum Alloys

STANDARDS
Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card
MS-28741	- Hose Assembly, Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (cont)

STANDARDS (cont)
Federal
Federal Test Method Standard No. 151 - Metals; Test Methods

DRAWINGS
Munitions Command

D8593745	- Cartridge, Aircraft Ejection Seat, Catapult, M37, Assembly
F8797379	- Carton, Packing Ammunition for Cartridge, Aircraft Ejection Seat, Catapult, M37
F8797380	- Box, Packing, Ammunition for Cartridge, Aircraft Ejection Seat, Catapult, M37
IEL-8593745	- Index to Inspection Equipment List for Cartridge, Aircraft Ejection Seat, Catapult, M37

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawing and referenced specifications.

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Vibration.- The assembled cartridge shall withstand vibration prescribed in MIL-E-5272, procedure XII without evidence of any disintegration of primer, igniter or propellant or derangement of components.

3.4 Physical properties.- Physical properties required by the drawings shall be uniform throughout the part within specified tolerances. Requirements for all alloy steel parts shall be achieved by quenching and tempering.

3.5 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.6 Functional.

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3.6.1 The cartridge shall not fail to fire.

3.6.2 Ballistics.- The cartridge when assembled in an M4A1 catapult shall propel a 325 pound mass vertically and shall comply with the requirements of Table I.

TABLE I

Temperature Degree(°) Fahrenheit(F)	Minimum Velocity ft/sec	Max Acceleration g's Pressure Time	Max Acceleration g's Load Cell	Max Rate of Change of Acceleration g/sec	Ignition Delay Milliseconds (See 6.1)
-65° to -70°F	33	12	10	85	150 max
65° to 75°F	38	12.5	10.5	100	150 max
160° to 165°F	41	14	14	125	150 max

3.6.2.1 Initiation.- The catapult shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 10 feet of aircraft hose, size 9, Standard MS-28741 at 70° ± 5°F.

3.7 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.7.1 Metal defects.- The part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.7.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure suppliers and services conform to prescribed requirements.

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4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production a sample shall be submitted in accordance with contract requirements and shall consist of 35 assembled cartridges, 50 cartridge cases for hardness test and 10 sets of unassembled components (including explosive components) for visual and dimensional examination and shall be tested as indicated in Table II herein.

TABLE II

Req	Test			Type of Test Dest	Non-Dest
3.3	4.4.1	Vibration	12 (from ballistic sample)	-	X
Dwg	4.4.2	Airtightness	Each	-	X
3.4	4.4.3	Hardness	50	-	X
Dwg	4.4.5	Sealant application	5	X	-
3.6.2	4.4.6.1	Ballistics	30	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and conform to requirements as indicated in Table I. The sample for firing shall consist of the following:

TABLE III

Temperature	Not Vibrated	Vibrated
-65°F to -70°F	6	4
65°F to 75°F	6	4
160°F to 165°F	6	4

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4.2.1.3 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, a lot shall consist of 5000 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process.
b. Propellant from not more than 1 lot.
c. Primers from not more than 1 lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class the criteria in Table V shall be used. All nonconforming material shall be rejected. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639, shall be used as a basis of comparison for surface roughness determination. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1 - 300	100%	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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TABLE V

Number of Defects in Class	Total Allowable Defects Per Sample Size Sample Size for Major Defects	110	150	225	300
2-3		1	2	3	4
4-7		2	3	4	5
8-19		3	4	5	7
20-43		4	5	8	10
over-43		6	8	11	14

Number of Defects in Class	Sample Size for Minor Defects	110	150	225	300
2-3		2	3	4	5
4-7		3	4	5	7
8-16		4	5	8	10
17-48		6	8	11	14
over-48		8	11	17	20

4.3.2.1 Primer, M42, dwg C7553919, covering a detail of dwg D8593745.

Defect	Method of Inspection
CRITICAL	
1. Improper assembly	X-ray[1/] and visual
2. Any part missing	X-ray[1/] and visual
3. Primer pellet broken, cracked or dislodged	X-ray[1/] and visual
4. Metal defective	X-ray[1/] and visual
5. Primer foil discolored	Visual
6. Mixed primers	Visual
MAJOR	
101. Burr	Visual
102. Foreign matter	Visual
MINOR	
None defined.	

1/ Prior to assembly into a cartridge, the primer shall be X-rayed with the sides of the cup perpendicular to the rays. The X-ray negatives, properly identified, shall be forwarded to the procuring activity.

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4.3.2.2 Sealing Disc, dwg C8593494, covering a detail of dwg D8593745.

	Defect	Method of Inspection
CRITICAL		
	None defined.	
MAJOR		
101.	Diameter, min.	Gage
102.	Marking missing, incorrect or illegible	Visual
103.	Metal defective	Visual
104.	Protective finish inadequate or defective	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
201.	Diameter, max	Gage
202.	Thickness	Gage
203.	Varnish missing	Visual

4.3.2.3 Retaining Ring, dwg B8593499, covering a detail of dwg D8593745.

CRITICAL		
	None defined.	
MAJOR		
101.	Concentricity of center hole with body	Gage
102.	Diameter of body, min	Gage
103.	Metal defective	Visual
104.	Protective finish	Visual
105.	Burr	Visual
106.	Foreign matter	Visual
MINOR		
201.	Diameter of center hole	Gage
202.	Diameter of body, max	Gage
203.	Thickness	Gage

4.3.2.4 Plug, Firing, dwg 8593498, covering a detail of dwg D8593745.

CRITICAL		
1.	Length of striker tip, min	Gage

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4.3.2.4 (cont)

Defect	Method of Inspection
MAJOR	
101. Diameter of body, min	Gage
102. Profile of striker tip	Gage
103. Height of boss	Gage
104. Metal defective	Visual
105. Burr	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of body, max	Gage
202. Diameter of boss	Gage
203. Total length	Gage
204. Length of striker tip, max	Gage
205. Surface finish improper	Visual

4.3.2.5 Igniting Disc, dwg B8593627, covering a detail of dwg B8593745.

CRITICAL	
None defined.	
MAJOR	
101. Diameter, min	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter, max	Gage
202. Thickness	Gage

4.3.2.6 Retaining Spring, dwg B8593629, covering a detail of dwg D8593745.

CRITICAL	
None defined.	
MAJOR	
101. Thickness of wire, min	Gage
102. Metal defective	Visual
103. Protective finish inadequate or defective	Visual

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4.3.2.6 (cont)

Defect	Method of Inspection
MAJOR (cont)	
104. Burr	Visual
105. Foreign matter	Visual
MINOR	
201. Diameter free	Gage
202. Thickness of wire, max	Gage
203. Gap, min (when spring is compressed to specified OD)	Gage

4.3.2.7 Cartridge Case, dwg C8593746, covering a detail of dwg D8593745.

CRITICAL	
None defined.	
MAJOR	
101. Concentricity of head with body	Gage
102. Straightness of base	Gage
103. Diameter of body, min	Gage
104. Thickness of base, min	Gage
105. Thickness of sidewall, min	Gage
106. Diameter of head, min	Gage
107. Metal defective	Visual
108. Protective finish inadequate or defective	Visual
109. Burr	Visual
110. Foreign matter	Visual
MINOR	
201. Diameter of body, max	Gage
202. Diameter of head, max	Gage
203. Total length	Gage
204. Thickness of sidewall, max	Gage
205. Location of datum diameter	Gage
206. Radius or chamfer missing or incorrect	Visual
207. Surface finish improper	Visual

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4.3.2.8 Cartridge Head, dwg C8593747, covering a detail of dwg D8593745.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
101. Concentricity of primer pocket with pocket recess and body	Gage
102. Concentricity of recess on top of head with primer pocket	Gage
103. Concentricity of flange on head with body	Gage
104. Diameter of primer pocket	Gage
105. Diameter of primer pocket recess, min	Gage
106. Diameter of body, min	Gage
107. Diameter of flange, min	Gage
108. Depth of flange, min	Gage
109. Depth of primer pocket to chamfer at base	Gage
110. Depth to bottom of primer pocket recess	Gage
111. Thickness of wall, min	Gage
112. Location of datum diameter	Gage
113. Metal defective	Visual
114. Protective finish inadequate or defective	Visual
115. Surface finish of primer pocket, primer pocket recess and top of head	Visual
116. Burr	Visual
117. Foreign matter	Visual
MINOR	
201. Diameter of primer pocket recess, max	Gage
202. Diameter of recess on top of head	Gage
203. Diameter of center hole	Gage
204. Diameter across center boss	Gage
205. Diameter across cut out	Gage
206. Diameter across spring retaining groove	Gage
207. Diameter of body, max	Gage
208. Diameter of step at spring retainer groove	Gage
209. Diameter of flange, max	Gage
210. Depth of recess on top of head	Gage

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4.3.2.8 (Cont'd)

Defect	Method of Inspection
MINOR (cont'd)	
211. Depth of cut-out	Gage
212. Depth of lip of cut-out	Gage
213. Depth to lip cut-out	Gage
214. Total length	Gage
215. Width of undercut	Gage
216. Location of center holes	Gage
217. Radius or chamfer missing or incorrect	Visual
218. Surface finish improper, except as otherwise classified	Visual

4.3.2.9 Cartridge, Aircraft Ejection Seat, Catapult, M37 Assembly, dwg. D8593745.

CRITICAL	
1. Primer not properly seated	X-ray[1/] and visual
2. Any part missing	X-ray[1/] and visual
3. Improper assembly	X-ray[1/] and visual
4. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Crimp missing or improper	Visual
103. Sealant missing	Visual
104. Sealant in cavity	Visual
105. "O" ring split, cracked or cut	Visual
106. Foreign matter	Visual
MINOR	
201. Diameter of body, max	Gage
202. Diameter of head, max	Gage
203. Total length, max	Gage
204. Location of datum diameter	Gage
205. Varnish missing or inadequate	Visual
206. Fold, wrinkle or bulge	Visual

1/ The cartridge shall be X-rayed with the long axis of the cartridge perpendicular to the rays of the machine. The X-ray negatives, properly identified, shall be forwarded to the procuring activity.

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4.3.2.10 Carton, Packing, Ammunition for Cartridge Aircraft Ejection Seat, Catapult, M37, dwg F8797379.

Defect	Method of Inspection
CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual and Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incomplete or illegible	Visual

4.3.2.11 Box, Packing, Ammunition for Cartridge, Aircraft Ejection Seat, Catapult, M37, dwg F8797380.

CRITICAL	
None defined.	
MAJOR	
None defined.	
MINOR	
201. Contents loose	Manual and Visual
202. Hardward missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

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4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Vibration.- Twelve (12) of the 30 cartridges of the initial production sample shall be subjected to vibration. The 12 cartridges shall then be X-rayed. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. Any change in the process of a lot of primers in subsequent production shall be reason to select an additional sample of 12 cartridges for vibration tests.

4.3.3.2 Airtightness.- Each cartridge shall be subjected to the specified pressure differential. Failure to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.3 Hardness.- Fifty (50) units from each heat treated lot of components shall be tested. Failure of any unit to comply with the requirements of 3.4 shall be cause for rejection of the lot of components.

4.3.3.4 Protective finish.- Twenty-three (23) units of each component shall be selected for protective finish tests. Failure of any component to comply with the requirements of 3.5 shall be cause for rejection of the lot.

4.3.3.5 Sealant application.- Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot of cartridges.

4.3.3.6 Functional.- The following test shall be performed by the Government:

4.3.3.6.1 Ballistics.- Thirty (30) assembled cartridges from the initial production sample and 30 cartridges from each subsequent production lot shall be grouped and temperature conditioned as prescribed in Table VI and 4.4.6.1.1 and functionally tested. Failure of any cartridge to comply with the requirements of 3.6.2 shall be cause for rejection of the lot.

TABLE VI

Temperature	Sample Size
-65°F to -70°F	10
65°F to 75°F	10
160°F to 165°F	10

4.3.4 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8593745.

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4.4 Test methods and procedures.

4.4.1 Vibration.- The method and procedure for performing the vibration test shall be in accordance with Specification MIL-E-5272, Procedure XII.

4.4.2 Airtightness.- The method for determining airtightness shall be by use of equipment listed on IEL-8593745.

4.4.3 Hardness.- Prior to the application of protective finish, a hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.4.4 Protective finish.- Test procedures for protective finish shall be in accordance with Federal Test Method Standard No. 151. The cartridge case and cartridge head shall be sectioned longitudinally to better expose the significant surfaces to the salt spray.

4.4.5 Sealant application.- The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the full length of the joining surfaces at the junction of the sealing disc and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Functional.

4.4.6.1 Ballistics.- The cartridge shall be loaded into an acceptable M4A1 Catapult, divided into 3 equal groups and temperature conditioned in accordance with Table I. The cartridge shall be tested within 10 minutes after removal from the conditioning chamber. When this time is exceeded the catapult shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning.- A standard M4A1 Catapult loaded with a dummy M37 cartridge, containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a mimimum of 1 hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

5. PREPARATION FOR DELIVERY.

5.1 Packing and marking.- Packing and marking shall be in accordance with drawing F8797379 and F8797380.

5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

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6. NOTES

6.1 Ignition delay is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

Custodian:

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Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on *the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and* at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT

$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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